Review notes · for you

How attacks work, and how to defend: unit check

Cyber security · Level 3 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Malware: how it gets in
- Social engineering: hacking people
- Attacks on passwords, and the defences
- Secret messages: encryption basics

Activity: Defender's map

1. Draw (5 min): Draw your phone in the middle of a page. Around it, draw four arrows for the four attacks: malware, social engineering, password attacks and stolen keys.
2. Defend (8 min): On each arrow, write the defences that block it: official store, updates and backups; pause and check another way; long unique passphrases and two-step sign-in; keep codes and PINs secret.
3. Check (4 min): Tick every defence you already have on your own phone and accounts.
4. Plan (3 min): Circle one defence you don't have yet, and write when you'll set it up.

Keep the map: it's a checklist you can show someone at home.

Discussion

- Which attack in this unit is most likely to reach you, and why?
- Which defence protects against more than one attack?
- Who at home would you most like to teach, and what?

Check yourself

- The one defence I'll set up this week, and when.

Try it this week
Pick the defence from this course that you need most (updates, a backup, unique passphrases or two-step sign-in), set it up this week, and teach it to one person at home.

Quiz sheet

How attacks work, and how to defend: unit check

Cyber security · Level 3 · Unit check

Name _____ Class _____ Date _____

1 **Which kind of malware pretends to be something useful, like a free game hack, so you install it yourself?**

◯ A worm ◯ Spyware ◯ A trojan ◯ Ransomware

2 **If you think your phone is infected, turning off Wi-Fi and mobile data is a good first step.**

◯ True ◯ False

3 **A caller says: "I'm from the bank's security team. Your account will be frozen in 15 minutes unless you read me the code we just sent." Which tricks is the caller using?**

☐ A gift ☐ Authority ☐ Fear ☐ Urgency ☐ Curiosity

4 **Put the steps for cracking a long Caesar message by counting letters in order.**

☐ Find the most common letter

☐ Count how often each letter appears

☐ Shift every letter back and check it reads

☐ Work out the shift from it to E (or alif)

5 **An attacker with a stolen password file can guess about ten billion times a second. About how long would it take to try every password made of eight random lower-case letters?**

◯ About 20 seconds ◯ About 20 days ◯ About 90 years

6 **Match each danger to the defence that stops it.**

A modded app from a chat group	• • • •	A different password for every account
Ransomware locks your files	• • • •	Install only from the official store
Credential stuffing	• • • •	Hang up and call a number you know
A fake "IT support" call	• • • •	A backup kept unplugged

7 **Complete the sentences.**

Encryption scrambles a message so that only someone with the _____ can read it. Caesar's cipher has so few keys that you can _____ them all. Modern encryption is so strong that attackers go after the key by _____ people instead.

Word bank: try · app · hide · helping · key · tricking

8 **What best protects your schoolwork from ransomware?**

◯ A longer screen-lock PIN ◯ Being ready to pay quickly ◯ A recent backup kept unplugged or in the cloud

9 **After a password file leaks, a long random passphrase still protects you far better than a short, complicated password.**

◯ True ◯ False

10 **In a long English message written with Caesar's cipher, the most common letter is H. E is the most common letter in English. What is the most likely shift?**

◯ 1 ◯ 8 ◯ 3

Check your answers

How attacks work, and how to defend: unit check

Cyber security · Level 3 · Unit check

1 **A trojan**

A trojan is named after the wooden horse: a gift that hides something harmful, so you carry it in yourself.

2 **True**

Cutting the connection stops malware spreading to other devices or sending your data out.

3 **Authority; Urgency; Fear**

"The bank's security team" claims authority, "15 minutes" is urgency, and "frozen" is fear.

4 **Top to bottom: 2, 1, 4, 3**

This is Al-Kindi's method: the most common letter in the secret message probably stands for the most common letter in the language, which reveals the shift.

5 **About 20 seconds**

Each of the eight letters has 26 choices, so there are eight 26s multiplied together (26 to the power of 8): about 209 billion possibilities.

6 **A modded app from a chat group — Install only from the official store**
Ransomware locks your files — A backup kept unplugged
Credential stuffing — A different password for every account
A fake "IT support" call — Hang up and call a number you know

Every attack in this course has a defence that closes its door.

7 **Encryption scrambles a message so that only someone with the key can read it. Caesar's cipher has so few keys that you can try them all. Modern encryption is so strong that attackers go after the key by tricking people instead.**

The key is the secret. Caesar's 25 or 27 keys can all be tried, but modern keys can't, so attackers trick people into handing over passwords, codes and PINs instead.

8 **A recent backup kept unplugged or in the cloud**

With a recent backup you can restore your work without paying anyone.

9 **True**

Each extra random word multiplies the attacker's work by thousands, while symbol swaps in a short password are among the first things tried.

10 **3**

E is the 5th letter and H the 8th, so the shift is the distance between them: 3.